Supplemental Material for "Amplitude-Robust Qubit Spectroscopy Using Echo-Root-Lorentzian Pulse Shapes"

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This Supplemental Material provides the experimental setup and qubit coherence characterization; a precise definition of the finite echo-root-Lorentzian pulse and its resonant cancellation mechanism; the Bloch-equation coherence limit; pulse-cutoff and adiabaticity analyses; the numerical model and numerical protocol comparisons; and a dense pulse-length campaign with matched three-level linewidth comparisons. The AC-Stark discussion distinguishes the dressed-state shift and a phase-average estimate from the finite-pulse spectral center. The closing two sections examine pulse modifications—the midpoint phase inversion, DRAG quadrature, and cutoff-edge smoothing—and then extend the cutoff–duration study of the polynomial-corrected DRAG echo to the full cutoff range, including the linewidth–contrast tradeoff and numerical checks.

S1. MEASUREMENT SETUP

Experiments were performed on a QPU comprising 11 uncoupled superconducting qubits hosted in a dilution refrigerator. Control and readout were implemented with a Quantum Machines OPX1000 equipped with a microwave front-end module (MW-FEM), together with external cryogenic attenuators, filters, circulators, and amplifiers.

A schematic of the setup is shown in Fig. S1. The MW-FEM uses direct digital synthesis and digital upconversion: a programmed complex pulse envelope is translated to the configured microwave carrier and emitted directly from an RF output, without an external analog local oscillator or IQ mixer [1]. For acquisition, the returned microwave signal is digitally downconverted to a complex baseband stream. QUA demodulation then applies cosine and sine integration weights to the sampled signal and accumulates the result to obtain the measured quadratures [2]. Thus, separate MW-FEM outputs provide the qubit-drive and resonator-readout tones, while an MW-FEM input acquires the returning readout signal. A dedicated DC bias line was used to set the qubit to its flux sweet spot.

The device was mounted at the mixing-chamber stage (∼9 mK), with the drive and readout lines attenuated across the refrigerator stages. The returning readout signal passed through circulators and a 4 K high-electron-mobility transistor (HEMT) amplifier before entering the MW-FEM for digital downconversion and demodulation.

Table S1. Parameters of the primary qubit used in the experiment and numerical analysis. Transition frequencies are rounded to 1 MHz and coherence times to 0.1 μs.

Parameter	Value	Determination
f_{01}	4.267 GHz	Qubit spectroscopy
$f_{02}/2$	4.159 GHz	Two-photon spectroscopy
f_{12}	4.051 GHz	$2(f_{02}/2) - f_{01}$
$\alpha/(2\pi)$	−216 MHz	$f_{12} - f_{01}$
T_1	51.2 μs	Energy-relaxation fit
T_2^*	7.3 μs	Ramsey fit
T_2	16.4 μs	Hahn-echo fit
T_ϕ^*	7.9 μs	$[1/T_2^* - 1/(2T_1)]^{-1}$
Γ_{f,T_2^*}	43.5 kHz	$1/(\pi T_2^*)$

S1.1. Qubit parameters

Table S1 summarizes the transition frequencies and coherence parameters of the primary qubit. The additional high-drive and dense pulse-length measurements use a second qubit, with the corresponding parameters specified alongside those measurements. The T_1 and Ramsey T_2^* values were obtained from exponential and exponentially damped sinusoidal fits, respectively. The Hahn-echo T_2 was obtained from an echo-decay fit. Calculations use the Ramsey value as a conservative coherence reference, $T_{2,\mathrm{eff}} = T_2^* = 7.3$ μs.

S2. FINITE PULSE DEFINITION AND ECHO CANCELLATION

We define the finite generalized Lorentzian envelope by

$$f_{n,c,L}(t) = \begin{cases} \left[1 + (t/\tau)^2\right]^{-n}, & |t| \le L/2, \\ 0, & |t| > L/2, \end{cases} \tag{S1}$$

where L is the total pulse duration, n is the shape exponent, and the relative endpoint amplitude is $c = f_{n,c,L}(L/2)$. These parameters imply

$$\tau = \frac{L/2}{\sqrt{c^{-1/n} - 1}}. \tag{S2}$$

The root-Lorentzian pulse used here corresponds to $n = 1/2$. Its echo-root-Lorentzian version (abbreviated as

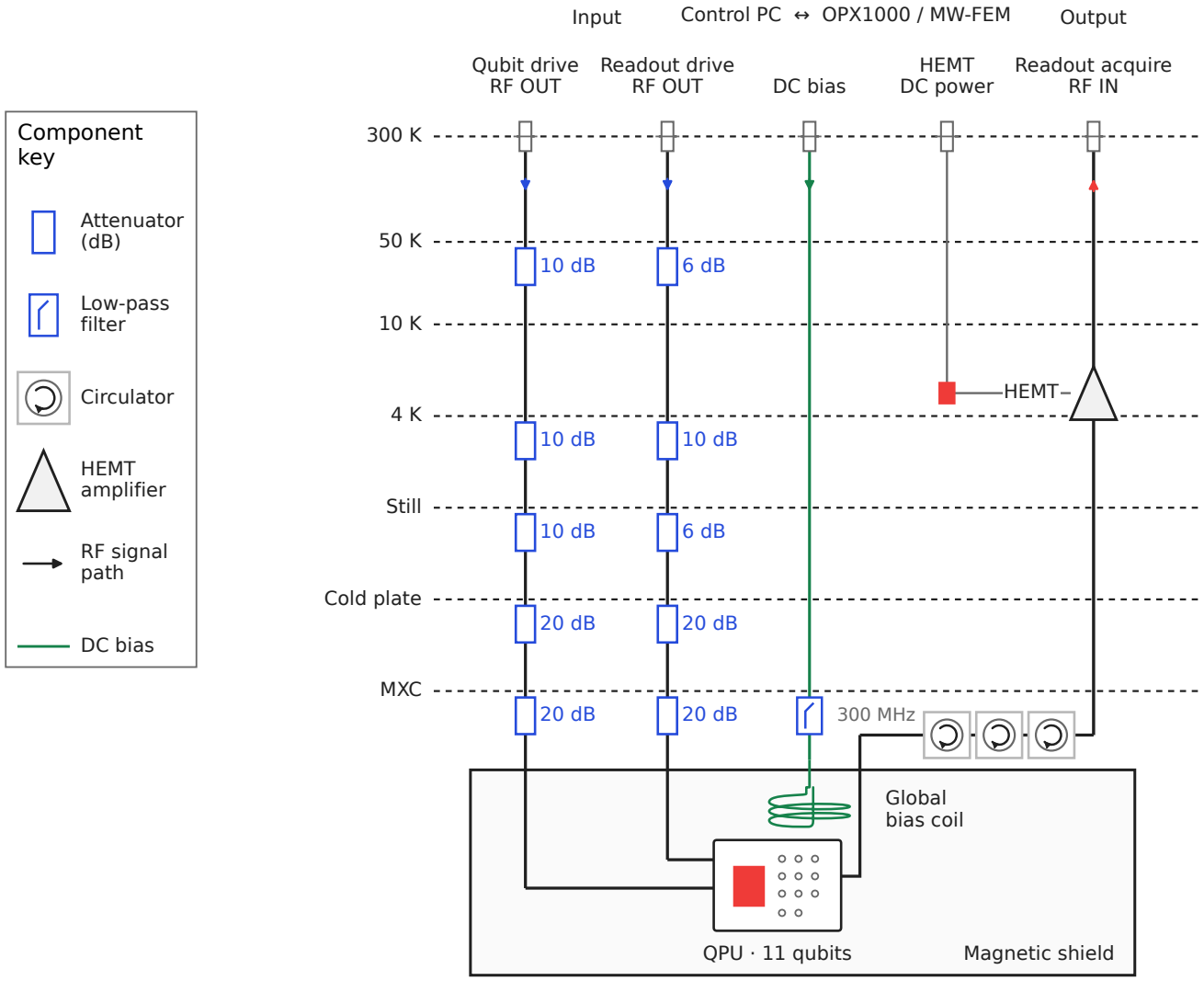


Figure S1. Schematic of the experimental setup for qubit control and readout. A control computer submits QUA programs and receives data over Ethernet. The OPX1000 MW-FEM directly generates the qubit-drive and resonator-readout RF signals and acquires the returning readout signal. Input lines are attenuated across the 300 K, 50 K, 10 K, 4 K, still, cold-plate, and mixing-chamber stages. At the mixing chamber, the device is isolated by circulators; the returning readout signal is amplified by a 4 K HEMT before entering the MW-FEM for digital downconversion and demodulation.

echo) has a phase inversion at the center,

$$\Omega_{\mathrm{E}}(t) = \Omega_0 f_{n,c,L}(t) s(t), \qquad s(t) = \begin{cases} +1, & t < 0, \\ -1, & t \geq 0. \end{cases} \tag{S3}$$

The value assigned at the single point $t = 0$ has no effect in the continuum limit. In a sampled waveform, the two halves are constructed with equal duration and opposite phase.

Two cutoff scans must be distinguished. A pure truncation scan keeps τ, n, and Ω_0 fixed while changing L. A fixed-duration scan instead keeps L fixed and changes the endpoint ratio c; in the parametrization of Eq. (S2), this also changes τ. Results from these two scans need not coincide.

In the rotating-wave approximation, the unitary two-level Hamiltonian is

$$\frac{H(t)}{\hbar} = \frac{1}{2}\left[\Delta\sigma_z + \Omega_{\mathrm{E}}(t)\sigma_x\right]. \tag{S4}$$

At exact resonance, Hamiltonians at different times are proportional to the same operator and therefore com-

mute. The propagator is

$$U(\Delta = 0) = \exp\left[-\frac{i\sigma_x}{2}\int_{-L/2}^{L/2}\Omega_{\mathrm{E}}(t)\,dt\right] = \mathbb{I}, \qquad \text{(S5)}$$

because the signed areas of the two symmetric halves cancel. For $\Delta \neq 0$, the σ_z and σ_x terms do not commute, so the cancellation is incomplete and a narrow detuning-dependent response remains. Equation (S5) is exact only for a symmetric, unitary two-level pulse. Relaxation, dephasing, phase or amplitude imbalance, and finite waveform bandwidth prevent exact cancellation in the experiment.

S3. BLOCH-EQUATION COHERENCE LIMIT

For comparison with the shaped-pulse protocol, we first summarize the constant-drive reference: the steady-state response of a continuously driven two-level system. We define $w \equiv \rho_{00} - \rho_{11}$, with $|0\rangle$ the ground state and $|1\rangle$ the excited state. Hence $P_e = \rho_{11} = (1 - w)/2$; no Pauli matrix sign convention is left implicit. Solving the optical Bloch equations for a drive with angular Rabi frequency Ω_0, angular detuning Δ, relaxation time T_1, and transverse coherence time T_2 gives [3]

$$\begin{aligned} w_{\mathrm{ss}}(\Delta) &= \frac{1 + \Delta^2 T_2^2}{1 + \Delta^2 T_2^2 + \Omega_0^2 T_1 T_2}, \\ P_{e,\mathrm{ss}}(\Delta) &= \frac{1}{2}\frac{s}{1 + s + \Delta^2 T_2^2}, \quad s \equiv \Omega_0^2 T_1 T_2. \end{aligned} \qquad \text{(S6)}$$

This expression assumes a pulse long enough to approach steady state. It is a reference result for conventional saturation spectroscopy, rather than a closed-form description of the finite echo-root-Lorentzian pulse.

At resonance, $P_{e,\mathrm{ss}}(0) = s/[2(1+s)]$, which approaches the saturated value $1/2$ only for $s \gg 1$. The positive half-maximum detuning is the angular-frequency HWHM

$$\Delta_{\mathrm{HWHM}} = \frac{1}{T_2}\sqrt{1 + s}. \qquad \text{(S7)}$$

The angular-frequency FWHM is therefore

$$\Gamma_\omega \equiv 2\Delta_{\mathrm{HWHM}} = \frac{2}{T_2}\sqrt{1 + s}. \qquad \text{(S8)}$$

The corresponding cyclic-frequency FWHM plotted in the Letter is

$$\Gamma_f \equiv \frac{\Gamma_\omega}{2\pi} = \frac{1}{\pi T_2}\sqrt{1 + s}. \qquad \text{(S9)}$$

Thus Δ_{HWHM}, Γ_ω, and Γ_f are distinct quantities and are not denoted by a common symbol.

In the weak-drive limit $s \ll 1$,

$$\Gamma_{\omega,T_2} = \frac{2}{T_2}, \qquad \Gamma_{f,T_2} = \frac{1}{\pi T_2}, \qquad \text{(S10)}$$

which are the same coherence-limited FWHM expressed in angular frequency and hertz, respectively. In the strong-drive limit $s \gg 1$,

$$\Gamma_\omega \longrightarrow 2\Omega_0\sqrt{\frac{T_1}{T_2}}, \qquad \Gamma_f \longrightarrow 2f_R\sqrt{\frac{T_1}{T_2}}, \qquad \text{(S11)}$$

where $f_R = \Omega_0/(2\pi)$. These limits are governed by s, not by comparing Ω_0 with an ambiguously defined linewidth.

S4. PULSE TRUNCATION AND ENDPOINT AMPLITUDE

The ideal root-Lorentzian tail has infinite support, whereas the implemented pulse starts and ends at finite amplitude $\Omega_c = c\Omega_0$. Abruptly switching this residual drive produces an endpoint transient. A useful estimate of the associated population contribution is [4]

$$P_c(\Delta) \simeq \frac{\Omega_c^2}{\Omega_c^2 + \Delta^2}\left[1 - P_0(\Delta)\right], \qquad \Omega_c = c\Omega_0, \qquad \text{(S12)}$$

where P_0 denotes the response that would be obtained without the endpoint jump. The artifact becomes important when Ω_c is comparable to the linewidth scale of interest. Thus the cutoff cannot be assessed from c alone: it must be considered together with the peak drive and the desired resolution.

The Bloch-equation reference gives the dimensionless endpoint saturation parameter

$$s_c = \Omega_c^2 T_1 T_2. \qquad \text{(S13)}$$

Keeping the endpoint contribution in the weak-saturation regime requires

$$s_c \ll 1 \iff \Omega_c \ll \frac{1}{\sqrt{T_1 T_2}} \iff c \ll \frac{1}{\Omega_0\sqrt{T_1 T_2}}. \qquad \text{(S14)}$$

The inequality therefore bounds the endpoint amplitude from above. It is a perturbative design criterion, not a claim that the cutoff artifact has the steady-state Lorentzian lineshape.

S5. ADIABATIC-BASIS DESCRIPTION

For one half of the pulse, the rotating-frame Hamiltonian can be written

$$\frac{H(t)}{\hbar} = \frac{1}{2}\begin{bmatrix} \Delta & \Omega(t) \\ \Omega(t) & -\Delta \end{bmatrix}. \qquad \text{(S15)}$$

Define

$$\epsilon(t) = \sqrt{\Delta^2 + \Omega^2(t)}, \qquad \vartheta(t) = \frac{1}{2}\,\mathrm{atan2}[\Omega(t), \Delta]. \qquad \text{(S16)}$$

Both ϵ and $\dot{\vartheta}$ have units of inverse seconds, whereas ϑ is dimensionless. The instantaneous eigenvalues are $E_- = -\hbar\epsilon/2$ and $E_+ = +\hbar\epsilon/2$. Taking the instantaneous eigenvectors as the columns of

$$R(t) = \begin{bmatrix} -\sin\vartheta & \cos\vartheta \\ \cos\vartheta & \sin\vartheta \end{bmatrix}, \tag{S17}$$

the transformed Hamiltonian is $H_a = R^{\mathsf{T}} H R - i\hbar R^{\mathsf{T}}\dot{R}$. Therefore

$$\frac{H_a(t)}{\hbar} = \begin{bmatrix} -\epsilon/2 & -i\dot{\vartheta} \\ i\dot{\vartheta} & \epsilon/2 \end{bmatrix}, \qquad \dot{\vartheta} = \frac{\Delta\dot{\Omega}}{2(\Delta^2+\Omega^2)}. \tag{S18}$$

The nonadiabatic coupling is $|\dot{\vartheta}|$, while the instantaneous angular-frequency gap is ϵ. Adiabatic following therefore requires [5]

$$|\dot{\vartheta}| \ll \epsilon. \tag{S19}$$

Replacing the inequality by equality provides the local crossover estimate

$$\sqrt{\Delta_b^2 + \Omega^2(t_0)} = \frac{|\Delta_b \dot{\Omega}(t_0)|}{2[\Delta_b^2 + \Omega^2(t_0)]}, \tag{S20}$$

where t_0 is the time at which the nonadiabatic coupling is largest for the detuning under consideration. This is a scaling criterion, not an exact linewidth formula.

This smooth-basis construction applies separately to each half of the echo pulse; the imposed phase jump at $t = 0$ is a control discontinuity and is not represented by $\dot{\vartheta}$ in this local adiabatic estimate.

For $L^{(n)}(t) \sim |t|^{-2n}$, the inverse-power-tail derivation of Boradjiev and Vitanov requires $2n > 1$. For an untruncated generalized Lorentzian with $n > 1/2$, it gives

$$\Delta_b \tau \propto (\Omega_0 \tau)^{-1/(2n-1)}. \tag{S21}$$

As $n \to 1/2^+$, the exponent $1/(2n-1)$ diverges, indicating the strongest power-narrowing trend within the asymptotic family. The exact root-Lorentzian case $n = 1/2$ is nevertheless marginal and lies outside the domain of Eq. (S21); substituting $n = 1/2$ into that power law is invalid. In particular, the singularity does not imply zero FWHM. The finite pulse length, nonzero endpoint amplitude, decoherence, and echo phase inversion set the observed Γ_f, which is evaluated from the complete simulated and experimental response in this work.

S6. NUMERICAL MODEL

The unitary trajectories in main-text Fig. 1 solve the two-level Schrödinger equation with QuTiP's state-vector solver [6], recording populations at 1600 output times. They omit relaxation and dephasing; these output times are not the solver's internal integration steps.

The dissipative simulations use either two or three levels, as specified for each comparison below, with Markovian relaxation and dephasing. Main-text Figs. 2–4 use the three-state transmon model. In the frame rotating at the drive frequency, and within the rotating-wave approximation, the real-drive Hamiltonian is implemented as [7]

$$\frac{H(t)}{\hbar} = -\Delta a^\dagger a + \frac{\alpha}{2} a^{\dagger 2} a^2 + \frac{\Omega_0}{2} f(t)\left(a + a^\dagger\right), \tag{S22}$$

where $\Delta = \omega_d - \omega_{01}$ is the drive–qubit detuning, Ω_0 is the peak Rabi frequency, α is the anharmonicity, and a is the truncated oscillator lowering operator. In the effective two-level model, $a = |0\rangle\langle 1|$ and the anharmonicity term vanishes. In the two-level case, up to an irrelevant identity term, Eq. (S22) is equivalent to $H/\hbar = (\Delta/2)\sigma_z + (\Omega_0 f/2)\sigma_x$.

The unmodified pulses follow Eqs. (S1)–(S3); the smoothed envelopes and DRAG quadrature are defined in Sec. S10. The overall sign of the echo envelope is a phase convention and does not affect the measured populations.

The density matrix evolves according to the Lindblad master equation

$$\dot{\rho} = -\frac{i}{\hbar}[H(t), \rho] + \mathcal{D}\left[\sqrt{\frac{1}{T_1}}\, a\right]\rho + \mathcal{D}\left[\sqrt{\frac{2}{T_\phi}}\, a^\dagger a\right]\rho, \tag{S23}$$

where $\mathcal{D}[C]\rho = C\rho C^\dagger - \frac{1}{2}\{C^\dagger C, \rho\}$ and

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}. \tag{S24}$$

The primary-qubit calculations use $T_1 = 51.2$ μs and $T_{2,\mathrm{eff}} = 7.31$ μs, corresponding to $T_\phi = 7.87$ μs. The primary-qubit anharmonicity is inferred from repeated OPX1000 two-photon spectroscopy. With $f_{01} = 4.267106667$ GHz, three scans locate the sharp, interior response at $f_{02}/2 = 4.159106667$ GHz, including a 1 MHz-step confirmation scan. We therefore use $\alpha/(2\pi) = 2(f_{02}/2 - f_{01}) = -216.0$ MHz; the finest scan gives a grid-scale uncertainty of approximately 1.0 MHz for the inferred anharmonicity. Each calculation starts in $|0\rangle$, integrates Eq. (S23) over the complete pulse, and records the final state observables. Spectroscopy maps repeat this evolution over the detuning and drive grids specified for each comparison, using the corresponding pulse duration, cutoff, and coherence parameters.

Main-text Fig. 2 uses fourth-order Runge–Kutta integration of the three-level density matrix, with 12 800 uniform time steps in each 10 μs half of the 20 μs pulse ($\Delta t = 0.78125$ ns). The two halves are integrated separately so that no step crosses the abrupt sign reversal. Each simulated map has 126 drive values selected from the broad experimental grid and 401 detunings over either ± 25 or ± 0.5 MHz. Doubling the steps per half at

representative drive–detuning points changes P_e by at most 1.61×10^{-5} for the ordinary pulse and 4.62×10^{-6} for the echo. Figure S2 uses the same integrator at all 501 acquired detunings and the five selected experimental drive values, with its convergence check described in Sec. S7. These calculations retain the states $|g\rangle$, $|e\rangle$, and $|f\rangle$, with $\alpha/(2\pi) = -216.0$ MHz, and report $P_e = \rho_{ee}$ while retaining $P_f = \rho_{ff}$ in the archived figure data. The pulse-shaping calculations in main-text Figs. 3 and 4 use the same three-level Lindblad structure with complex drive $I - iQ$, including the derivative of the complete signed envelope. Figure 3 uses a 0.5 ns time step; Fig. 4 uses a maximum 0.25 ns step with a grid adapted to the pulse core. The latter scan covers five durations, 100 logarithmic cutoffs from 10^{-5} to 1, 100 drive amplitudes from 0 to 100 MHz, and 201 detunings over ±0.5 MHz. Main-text Fig. 4 shows selected slices of this scan. Widths spanning fewer than six detuning intervals, unresolved dips, and nonpositive contrast are rejected. Representative checks at three cutoffs and three amplitudes for each duration halve the maximum time step and double the detuning intervals; among mutually accepted traces, the largest FWHM changes are 0.255 kHz and 0.117 kHz, respectively. This is a sampled convergence check, not an error bound for every scan point. The three-level calculations in Sec. S10 likewise use fourth-order Runge–Kutta integration, with the time steps and refinement checks given alongside each comparison. The dense pulse-length calculation uses the sampled waveform and campaign-specific parameters described in Sec. S8.

The duration–resolution comparison in Fig. S3 instead integrates the two-level Bloch equations with fourth-order Runge–Kutta steps in $u = \operatorname{arsinh}(t/\sigma)$, where $\sigma = (L/2)/\sqrt{c^{-2}-1}$. The transformation $t = \sigma \sinh u$ gives $dt/du = \sigma \cosh u$ and concentrates physical-time samples near the pulse core. Each half uses 1600 uniform steps in u, with the sign reversal imposed between halves. The spectra are evaluated at 100 logarithmically spaced drives from 0.001 to 50 MHz and 201 detunings over ±0.5 MHz; the displayed drive range ends at 25 MHz. These calculations capture coherent drive dynamics, relaxation, and pure dephasing, but do not describe leakage, higher-level AC-Stark shifts, or multiphoton transitions. The main-text Figs. 2–4, fixed-amplitude spectra in Sec. S7, dense pulse-length comparison in Sec. S8, and high-drive analyses in Secs. S10–S11 use the explicit three-level model.

S7. ADDITIONAL SPECTRAL COMPARISONS

S7.1. Fixed-amplitude spectra

Figure S2 resolves the narrow-detuning maps of main-text Fig. 2 into five measured amplitude slices per protocol. The nearest acquired rows to nominal peak Rabi frequencies of 3, 10, 20, 30, and 40 MHz are selected without optimizing their agreement with the model. The three-level calculation uses the exact acquired drive values and all 501 measured detunings over ±0.5 MHz, with the same primary-qubit model parameters as main-text Fig. 2. Each measured point contains 2000 repetitions. Halving the integration step at all five drives and seven representative detunings changes P_e by less than 1.2×10^{-5}. This checks numerical accuracy independently of the model's agreement with experiment.

To compare line shapes independently of their absolute offsets, we subtract each trace's mean over the full displayed detuning interval and add the same stacking offset to its experimental and simulated members. The vertical scale is common to every trace: no contrast normalization, gain adjustment, frequency shift, or smoothing of the measured data is applied. This display retains differences in feature depth, width, and center, while removing only a constant population background. The unaligned maps remain in the main text.

S7.2. Duration–contrast tradeoff

Figure S3 quantifies the tradeoff between pulse duration, linewidth, and signal contrast at fixed cutoff $c = 0.005$. The continuous curves show the simulated response, while the markers show the experimental data. Panels (a) and (b) separate spectral narrowing from signal loss: longer pulses reach smaller FWHM relative to the T_2 limit, but their fitted contrast remains lower over the usable drive range. Their contrast-weighted operating windows in panel (c) therefore contract despite the high resolution. Pulse duration must consequently be selected jointly with contrast and experimental signal quality, rather than by linewidth alone.

S8. DENSE PULSE-LENGTH DEPENDENCE

S8.1. Acquisition and campaign-specific normalization

A separate campaign on the second qubit tests the dependence on the pulse duration L at fixed relative cutoff $c = 0.005$. Each duration $L = 1, 2, 5, 10, 15, 20, 25, 30, 35,$ and 40 μs was acquired once, using 400 uniformly spaced detunings over $[-1, 1]$ MHz and 400 logarithmically spaced nominal amplitudes from 0.01 to 0.70 V, with 380 averages per grid point. The waveform template duration equals L, with no duration stretching. The echo has an abrupt midpoint sign reversal; the derivative quadrature is disabled ($\beta = 0$), and no AC-Stark compensation is applied (effective $\kappa = 0$). The finite envelope therefore changes its time scale together with L, rather than varying the truncation time of a fixed-width pulse.

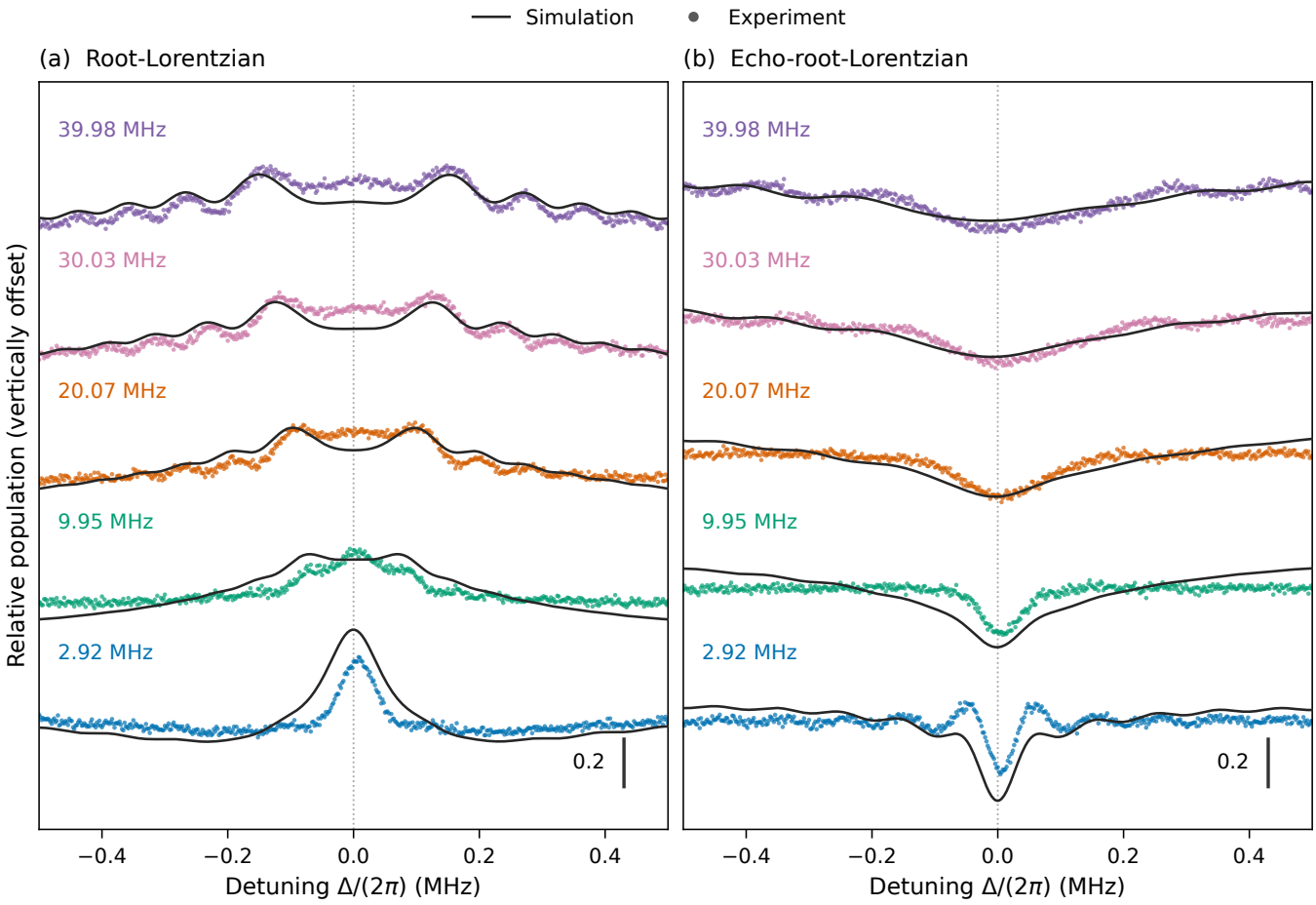


Figure S2. Stacked experimental and simulated spectra for (a) root-Lorentzian and (b) echo-root-Lorentzian pulses, with $L = 20$ μs and $c = 0.005$. From bottom to top, the acquired peak Rabi frequencies are approximately 2.92, 9.95, 20.07, 30.03, and 39.98 MHz, labeled individually. Colored markers show all measured points from the narrow-detuning scans in main-text Fig. 2; black lines show three-level simulations at the identical drive and detuning values. For each trace independently, $y = P - \langle P \rangle_\Delta + 0.55j$, where $j = 0, \ldots, 4$ indexes increasing drive. The scale bars denote a population difference of 0.2. Only constant vertical offsets are removed; contrast and frequency are not fitted. Two-state readout does not independently resolve $|f\rangle$, whereas the simulated observable is P_e.

The saved square-π calibration is $t_\pi = 40$ ns at $A_\pi = 0.203776$ V. We express the drive axis as the inferred nominal peak Rabi frequency,

$$f_R = \frac{\Omega_0}{2\pi} = \frac{A/A_\pi}{2t_\pi}, \tag{S25}$$

which gives 0.613–42.939 MHz for this sweep. This conversion uses the envelope scale, rather than its slightly smaller sampled maximum. The campaign records $T_1 = 41.712$ μs and $T_2^* = 6.122$ μs. Its linewidth reference is

$$\Gamma_{T_2^*} = \frac{1}{\pi T_2^*} = 51.996 \text{ kHz}. \tag{S26}$$

This campaign-specific Ramsey reference is distinct from the primary-qubit coherence normalization used elsewhere in the paper. It is a reference scale, not a universal lower bound on an echo-generated peak or dip.

S8.2. Linewidth extraction and matched simulation

For this campaign, linewidths are extracted with a direct half-height estimator. Each frequency trace is smoothed with a nine-point, second-order Savitzky–Golay filter. A linear baseline is fitted to the outer 10% at each scan edge; the strongest absolute residual feature is selected, and its nearest half-height crossings are linearly interpolated. Experimental widths are accepted only when both crossings lie inside the scan, the estimated contrast-to-noise ratio is at least five, and the width lies between three frequency bins (15.038 kHz) and 80% of the scan (1600 kHz). The smoothing window spans approximately 40 kHz and can affect narrow features. These selection rules do not provide calibrated linewidth uncertainties.

The matched calculation uses the three-level Lindblad

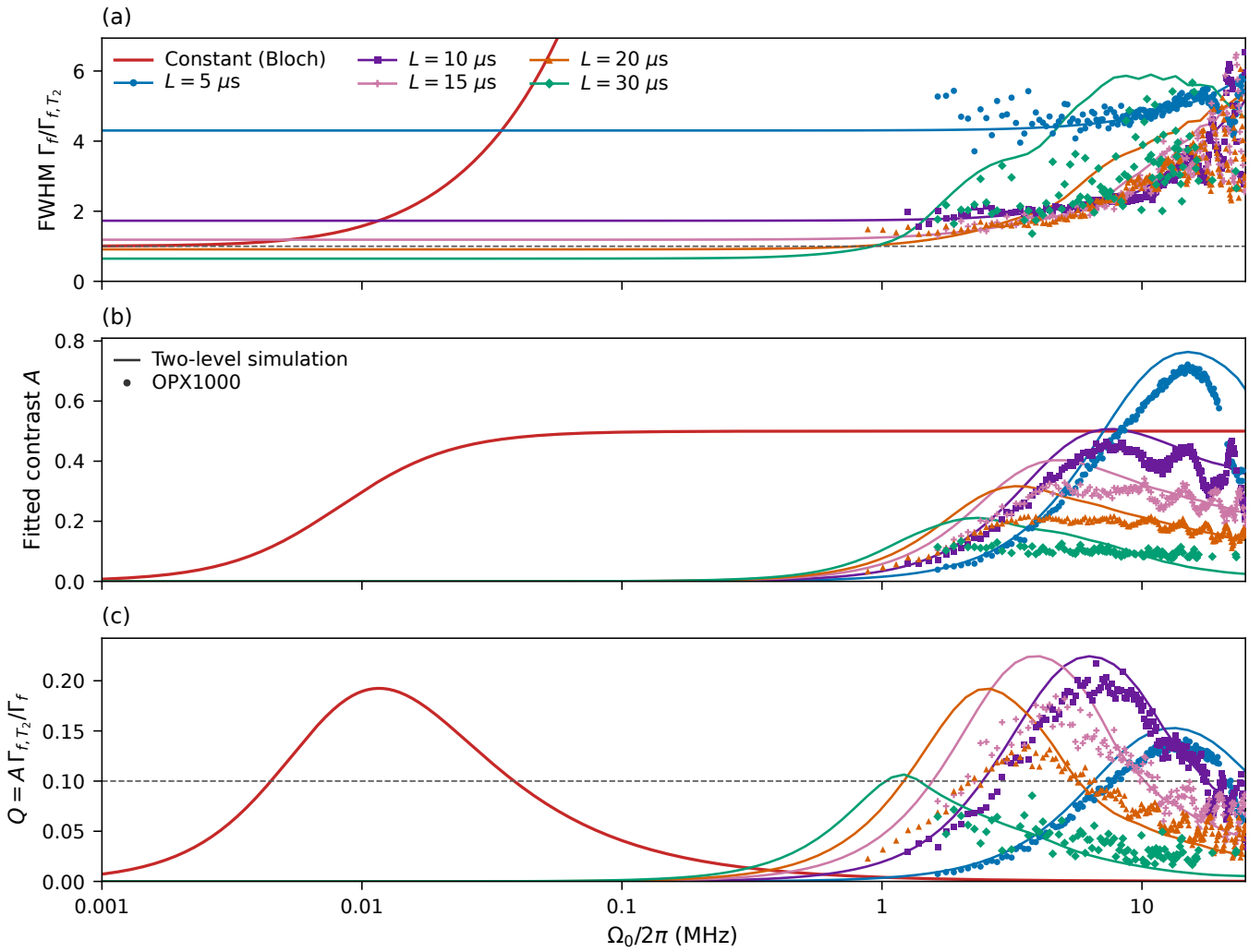


Figure S3. Resolution–contrast performance versus peak Rabi frequency up to 25 MHz for echo-root-Lorentzian pulses with $c = 0.005$ and $L = 5$, 10, 15, 20, and 30 μs. Colored continuous lines show two-level simulations, and colored markers show OPX1000 data. (a) Fitted FWHM in units of the coherence-limited reference, $\Gamma_f/\Gamma_{f,T_2}$, with the dashed line marking $\Gamma_f/\Gamma_{f,T_2} = 1$; (b) fitted contrast A; and (c) contrast-weighted resolution $Q = AR = A\Gamma_{f,T_2}/\Gamma_f$. The red curve is the constant-drive reference from the Bloch equations, and the dashed line in (c) marks $Q = 0.10$.

model of Sec. S6, with this campaign's saved coherence and pulse parameters and $\alpha/(2\pi) = -237.95$ MHz. It begins in $|0\rangle$, holds each waveform sample for 1 ns, and includes relaxation rates $\Gamma_{10} = 1/T_1$ and $\Gamma_{21} = 2/T_1$. Pure dephasing is Markovian, with rate $1/T_2^* - 1/(2T_1)$. The simulated observable is $P_1 + P_2$, assuming both excited levels are assigned the excited binary outcome. No parameters are fitted to the measured maps. The calculation omits readout errors, reset errors, drift, quasi-static noise and analog waveform distortion. The simulated linewidth uses the same half-height geometry, with contrast at least 0.001 in place of the experimental noise criterion. A reduced-grid comparison at every duration between 1 and 0.5 ns integration steps changes populations by less than 5.8×10^{-4}; this numerical check does not validate the physical model.

S8.3. Duration trends and experiment–simulation agreement

We compare experimental and simulated widths at the same acquired amplitude index and duration for $L = 1, 2, 5, 10, 15, 20$, and 25 μs, using the intersection of the two acceptance masks. There are 1278 pairs in total, with counts of 110, 135, 213, 260, 265, 263 and 32 for the seven increasing durations. At 2 μs the paired linewidth RMS difference is 34.2 kHz; it increases to 302.5 kHz at 20 μs. For most long-duration pairs, the simulation predicts broader features than the experiment. Pairing accepted widths does not by itself establish that the strongest feature selected in each trace has the same physical origin.

Only 32 of 400 experimental traces pass the selection

at 25 μs, and none do so at 30, 35 or 40 μs. These missing widths mean that the feature is not reliably resolved by this estimator, rather than that its linewidth vanishes. The surviving sets therefore differ across durations; the longer-duration comparison is conditional on the small accepted subset. Because the maps were acquired sequentially over approximately 8.6 hours without repeated campaigns, drift and pulse-length effects are not independently separated. The data support a duration-dependent narrowing and loss of measurable contrast, but do not establish an optimal duration or a universal coherence-limited plateau.

S9. THREE-LEVEL AC-STARK SHIFT OF THE f_{01} RESONANCE

We summarize the weak-drive AC-Stark scale of the f_{01} transition using the three-level rotating-wave model. The detuning convention is $\Delta = \omega_d - \omega_{01}$. The basis $\{|g\rangle, |e\rangle, |f\rangle\}$ retains $|f\rangle$ as a dynamical state; in the weak-drive interpretation, virtual $|e\rangle \leftrightarrow |f\rangle$ coupling produces the leading shift of f_{01}. In this basis, the rotating-frame Hamiltonian of Eq. (S22) is

$$\frac{H(t)}{\hbar} = \begin{pmatrix} 0 & \Omega(t)/2 & 0 \\ \Omega(t)/2 & -\Delta & \Omega(t)/\sqrt{2} \\ 0 & \Omega(t)/\sqrt{2} & -2\Delta + \alpha \end{pmatrix}, \tag{S27}$$

where $\alpha = \omega_{12} - \omega_{01} < 0$. For the constant-drive reference, virtual coupling between $|e\rangle$ and $|f\rangle$ gives the weak-drive result for a weakly anharmonic transmon [7, 8]:

$$\delta f_{01}^{(\mathrm{const})} \equiv \frac{\Delta_{\mathrm{res}}}{2\pi} \simeq -\frac{(\Omega_0/2\pi)^2}{2(\alpha/2\pi)}. \tag{S28}$$

Because $\alpha < 0$, the dressed f_{01} resonance of the constant-drive reference moves to positive detuning.

For the shaped pulses, $\Omega(t) = \Omega_0 f(t)$ with the root-Lorentzian envelope of Eq. (S1), order $n = 1/2$, and endpoint ratio c. A useful weak-drive phase-average estimate is

$$\begin{aligned} \delta f_{01}^{(\mathrm{shape})} &\simeq -\frac{(\Omega_0/2\pi)^2}{2(\alpha/2\pi)} \left\langle f^2 \right\rangle_t, \\ \left\langle f^2 \right\rangle_t &= \frac{2\sigma}{L} \tan^{-1}\left(\frac{L}{2\sigma}\right), \end{aligned} \tag{S29}$$

where $\sigma = (L/2)/\sqrt{c^{-2} - 1}$. The echo phase inversion changes $f(t) \to -f(t)$ in the second half and therefore does not reverse this quadratic Stark term. Consequently, the root-Lorentzian and echo-root-Lorentzian pulses have the same leading phase-average estimate.

The position of a peak or dip in the final population response need not coincide with this phase-average estimate: it also depends on finite-pulse interference and dissipation. Section S10 therefore evaluates the spectral center directly from the simulated response when comparing DRAG coefficients. The rotating-wave model omits the counter-rotating Bloch–Siegert contribution [9].

S9.1. AC-Stark scale at a common peak drive

To compare the shift of the constant-drive reference with the shaped-pulse phase-average estimate, take a common nominal peak drive $\Omega_0/(2\pi) = 50$ MHz. For the constant-drive reference, the weak-drive estimate from Eq. (S28) is approximately 5.8 MHz. For the 20 μs root-Lorentzian envelope with $c = 0.001$, multiplication by $\langle f^2 \rangle_t = 0.001570$ reduces the phase-average estimate to 9.1 kHz. These estimates compare the two envelopes at the same nominal peak drive using Eqs. (S28) and (S29). The echo phase inversion does not cancel the quadratic Stark term; rather, the small time-averaged squared envelope suppresses it, while finite-pulse interference determines the experimentally visible central minimum.

S10. PULSE MODIFICATIONS: PHASE INVERSION, DRAG, AND CUTOFF-EDGE SMOOTHING

The finite echo-root-Lorentzian pulse has two distinct sources of rapid waveform variation: the midpoint sign reversal and the turn-on and turn-off at the cutoff. We examine modifications of these features separately from the derivative quadrature used for DRAG. This distinction matters because smoothing the midpoint can suppress population transfer to $|f\rangle$ without removing the spectral effects of the outer edges, while DRAG alone need not narrow the central spectroscopy dip. The separate effects are examined with three-level simulations. The experimental comparison in Fig. S5(d) tests a combined change of midpoint smoothing and DRAG, rather than either modification in isolation.

Table S2 summarizes the role of each modification before the separate comparisons below.

S10.1. The midpoint phase jump and its smooth implementation

The ideal echo changes the carrier phase by π at the pulse midpoint, equivalently reversing the sign of its real envelope. In the numerical implementation we replace the abrupt sign change by

$$\begin{aligned} s_T(t) &= \begin{cases} +1, & t < -T/2, \\ -\sin(\pi t/T), & |t| \le T/2, \\ -1, & t > T/2, \end{cases} \\ I(t) &= \Omega_0 g(t) s_T(t), \end{aligned} \tag{S30}$$

Table S2. Pulse modifications and the limitations they address. The effects listed are established by the separate simulation comparisons below; the SNR is an ideal shot-noise estimate.

Modification	Problem and benefit
Midpoint smoothing	Abrupt sign reversal causes leakage into $\|f\rangle$. A finite sine transition suppresses final-state leakage and gives a finite derivative for DRAG (Fig. S4).
DRAG quadrature	Drive-induced phase error displaces the spectral center. Adding $Q \propto dI/dt$, with $\beta \simeq -0.5$ in our convention, reduces center motion; it does not by itself narrow the dip (Fig. S5).
Polynomial outer-edge smoothing	Hard truncation contributes to high-drive broadening and loss of visibility. Endpoint subtraction and the $p = 5$ window reduce broadening, improve contrast and ideal SNR, and maintain these improvements over a wider drive range at fixed DRAG (Fig. S7).

where $g(t)$ is the unsigned, even envelope and T is the transition duration. The $T = 0$ reference is an abrupt inversion. For $T > 0$, both s_T and its first derivative are continuous. This interpolation passes through zero in the real drive; it is not a constant-amplitude rotation of the carrier phase. Its odd symmetry preserves zero signed pulse area. Thus, for a real drive in the ideal resonant two-level model, the cancellation argument of Sec. S2 still applies. It does not imply exact cancellation in the dissipative three-level model or after adding a noncommuting quadrature.

The saved transition-duration comparison isolates this change with $L = 20$ μs, $c = 0.005$, $\beta = 0$, and no additional Stark-phase correction. It uses $T_1 = 51.24$ μs, $T_2 = 7.31$ μs, and $\alpha/(2\pi) = -216$ MHz. Figure S4 shows the final P_f over 201 detunings within ± 0.5 MHz at the highest drive, and its maximum over detuning for each of 25 drive scales from 0 to 42.72 MHz. The largest final populations on this grid are 3.19×10^{-2}, 7.23×10^{-3}, 4.65×10^{-4}, and 2.55×10^{-4} for $T = 0$, 4, 8, and 16 ns, respectively. Increasing T from 0 to 16 ns therefore reduces the largest residual leakage by about a factor of 125 on this grid. This is final-state leakage, not the maximum transient occupation of $|f\rangle$. Halving the integration step from 0.5 to 0.25 ns at representative drive–detuning points changes the final populations by at most 1.90×10^{-5} across the four transition durations. The subsequent matched waveform comparisons use the same 16 ns inversion with and without DRAG.

S10.2. DRAG acts on the complete signed waveform

Using the complex-drive convention $I - iQ$, the DRAG quadrature [10, 11] is

$$Q(t) = -\frac{\beta}{|\alpha|}\frac{dI}{dt} = -\frac{\beta\Omega_0}{|\alpha|}\left[g'(t)s_T(t) + g(t)s'_T(t)\right], \tag{S31}$$

where α is an angular frequency. Both the envelope derivative and the derivative of the midpoint inversion are included. Applying DRAG only to the unsigned envelope would omit the contribution where the signed drive changes most rapidly. A finite T regularizes that contribution, whereas the derivative of an ideal instantaneous sign change is singular.

The waveform and cutoff comparisons below use $\beta = -0.5$ in this convention, motivated by the phase-compensation condition and numerical scan that follow. No additional drive-phase correction is applied.

S10.3. Choice of the DRAG coefficient

The value $\beta = -0.5$ has a perturbative phase-compensation motivation. For a weakly anharmonic oscillator with the transmon matrix-element ratio $\sqrt{2}$, the leading-order half-derivative DRAG solution removes the phase error associated with the AC-Stark shift without an additional time-dependent detuning control [11]. In the quadrature convention used here, this gives

$$Q_{\mathrm{phase}}(t) \simeq -\frac{1}{2\alpha}\frac{dI}{dt} = \frac{1}{2|\alpha|}\frac{dI}{dt}, \qquad \beta_{\mathrm{phase}} = -\frac{1}{2}. \tag{S32}$$

This condition compensates the leading dynamical phase error; it does not make the instantaneous dressed AC-Stark shift vanish. The central minimum of a finite echo spectrum also depends on interference, dissipation, and the cutoff, so its optimal coefficient need not be exactly -0.5 or coincide with the minimum-leakage coefficient.

Figure S5 tests this choice for the 20 μs, $c = 0.005$, 16 ns sine-smoothed echo-root-Lorentzian pulse, using the same three-level parameters as the preceding leakage comparison. We scan 35 coefficients over $-1 \le \beta \le 0$, with spacing 0.005 near -0.5, and hold all other pulse parameters fixed. At seven equally spaced drive amplitudes from 10 to 42.72 MHz, the central minimum of P_e is located on a $[-60, 60]$ kHz detuning grid with 1 kHz spacing and refined by a parabola through the minimum and its two neighbors. The objective is the RMS center offset about zero detuning,

$$J(\beta) = \left[\frac{1}{7}\sum_{j=1}^{7}\left(\frac{\Delta_{\min,j}(\beta)}{2\pi}\right)^2\right]^{1/2}. \tag{S33}$$

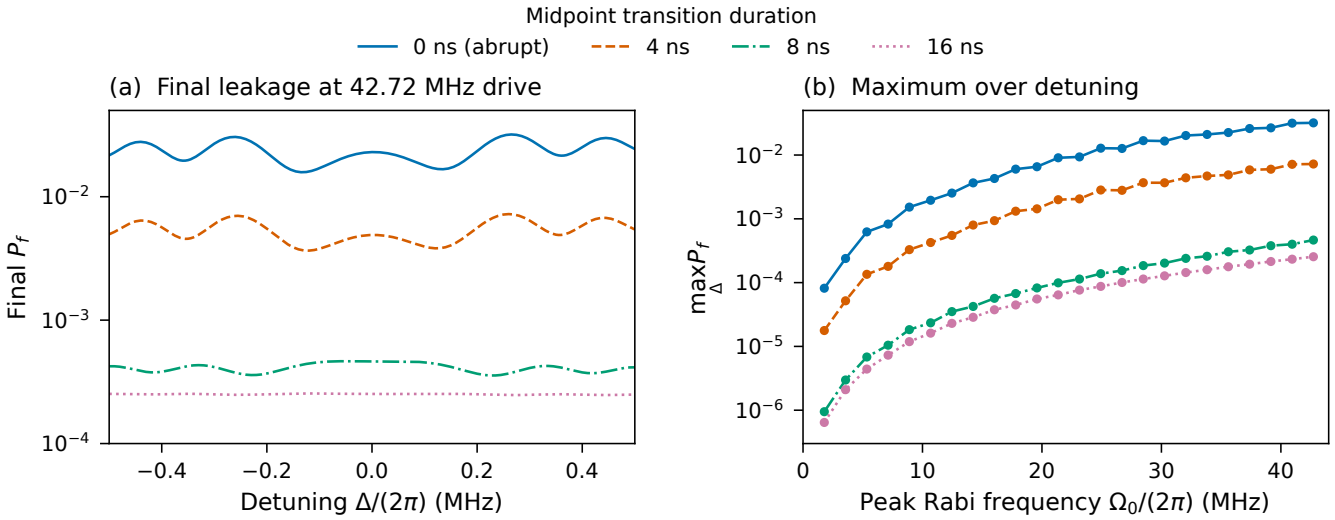


Figure S4. Simulated suppression of final-state leakage by smoothing the midpoint inversion of the echo-root-Lorentzian pulse. Curves compare $T = 0$ (abrupt), 4, 8, and 16 ns with $L = 20$ μs, $c = 0.005$, and $\beta = 0$ ($Q = 0$). (a) Final P_f versus detuning at $\Omega_0/(2\pi) = 42.72$ MHz. (b) Maximum final P_f over the 201 sampled detunings within ± 0.5 MHz, versus peak drive. Markers denote saved simulation amplitudes; lines connect samples without fitting or smoothing. Both leakage axes are logarithmic. The zero-drive point, for which $P_f = 0$ in every case, is omitted from (b). The three-level model and convergence checks are specified in the text; the plotted population is evaluated after the pulse and is not the maximum transient occupation of $|f\rangle$.

The smallest sampled value is $J = 0.099$ kHz at $\beta = -0.495$, compared with 0.125 kHz at -0.5 and 7.64 kHz at zero. Repeating the three coefficients -0.500, -0.495, and -0.490 with a 0.25 ns time step and 0.5 kHz detuning spacing preserves their ranking and changes individual centers by less than 0.10 Hz. Thus -0.5 is close to the numerical optimum for this center objective, rather than an exact universal optimum.

The experimental comparison in Fig. S5(d) compares two measured settings: $\beta = 0$ with an abrupt inversion and $\beta = -0.22$ with a 16 ns inversion, for $L = 50$ μs and $c = 0.00075$ on the second qubit. The coefficients are quoted in the acquisition convention. Each run's mean accepted center over the displayed 10–60 MHz range is subtracted; the RMS variations are 6.78 and 2.64 kHz, respectively. These measurements demonstrate improved center stability for the combined pulse change. Because both the quadrature and midpoint smoothing change, they neither isolate the effect of β nor establish an experimental optimum at -0.5.

S10.4. Smoothing the outer cutoff edges

For the root-Lorentzian core,

$$f(t) = \left[1 + (t/\sigma)^2\right]^{-1/2}, \qquad \sigma = \frac{L}{2}\frac{c}{\sqrt{1-c^2}}, \tag{S34}$$

hard truncation at $t = \pm L/2$ leaves an amplitude jump of size c. The polynomial correction first subtracts this endpoint value and then multiplies by a symmetric window:

$$g_p(t) = \frac{f(t) - c}{1 - c}\left[1 - \left(\frac{2t}{L}\right)^2\right]^p, \qquad |t| < L/2. \tag{S35}$$

It is zero outside the pulse. We use $p = 5$, which makes the envelope and its first five derivatives vanish at the boundaries. The corrected envelope retains $g_p(0) = 1$, is even, and uses the same nominal scale Ω_0. We apply neither pulse-area normalization nor renormalization to the peak of the signed, smoothed waveform. The cutoff c consequently specifies the underlying core width, not the nonzero endpoint amplitude of the corrected pulse. Figure S6 shows the envelope, turn-on edge, signed midpoint waveform, and DRAG quadrature for $L = 30$ μs and $c = 0.001$, corresponding to $\sigma \simeq 15$ ns. Even with smoother outer edges, the sine midpoint transition is only C^1 at its joins; outer-edge regularity should not be confused with the regularity of the entire signed pulse. The turn-off follows from the even symmetry of g_p. The signed $I(t)$ is odd, whereas its derivative and hence $Q(t)$ are even. The two contributions to $Q(t)$ in Eq. (S31) are shown separately: the midpoint term gs'_T produces the central correction, while $g's_T$ also contributes away from the inversion interval.

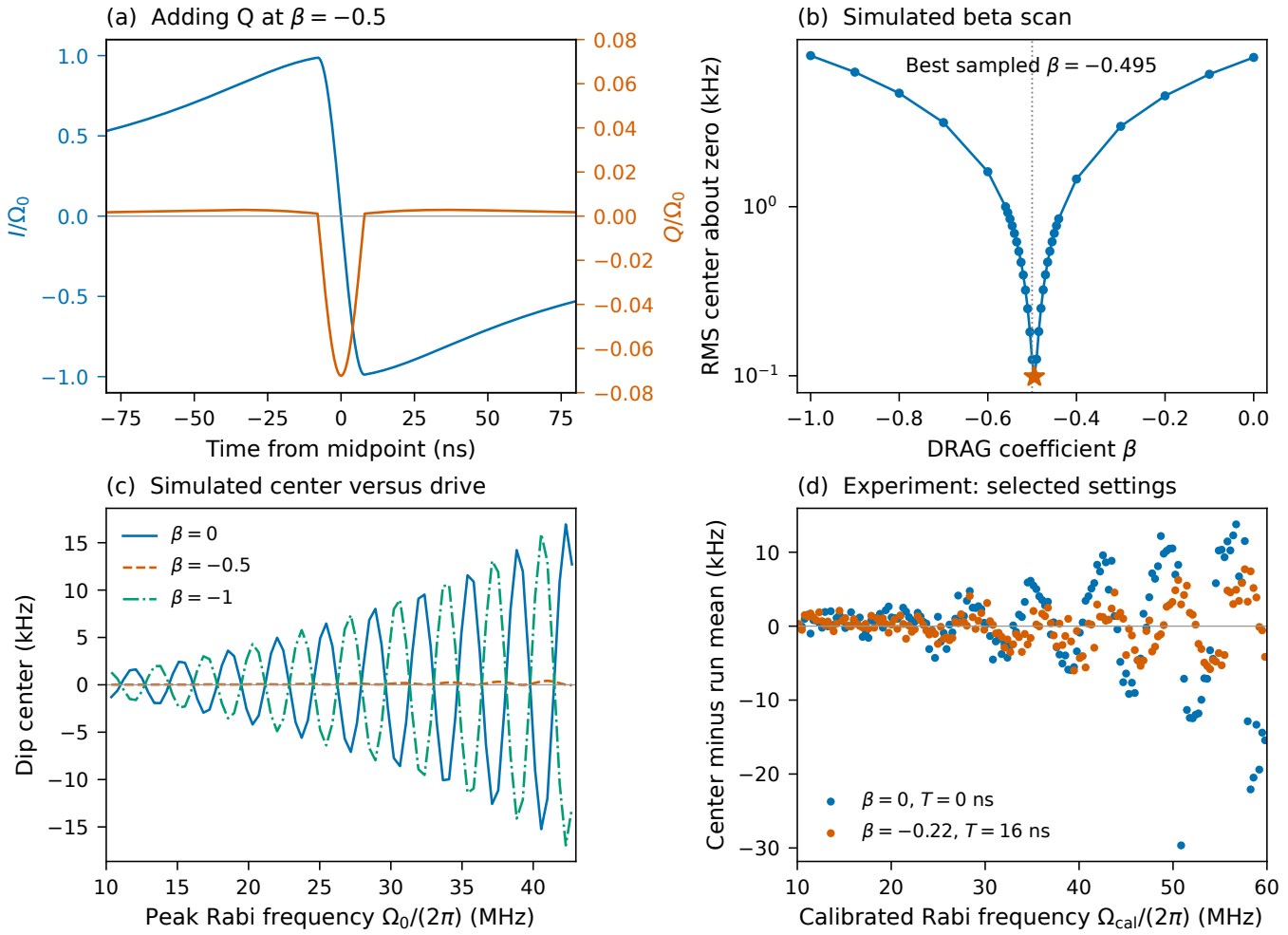


Figure S5. DRAG quadrature, simulated coefficient optimization, and experimental center stability. (a) Signed in-phase waveform (blue, left axis) and added quadrature (orange, right axis) at $\beta = -0.5$, with a 16 ns inversion interval. (b) Simulated RMS center offset $J(\beta)$; the star marks the lowest sampled value at -0.495, and the dotted line marks -0.5. (c) Direct central minima of the saved simulated spectra for $\beta = 0, -0.5, -1$; all three use identical midpoint smoothing. Panels (a)–(c) use $L = 20$ μs and $c = 0.005$ with no additional detuning correction. Lines connect sampled values. (d) Accepted Gaussian-fit centers from two measured 50 μs, $c = 0.00075$ pulses, with one mean subtracted per run and no additional smoothing. Each acquisition used 2000 shots per point. The settings differ in both β and inversion duration, as labeled; this panel is a measured comparison, not an experimental coefficient scan.

S10.5. Edge smoothing: linewidth, visibility, and drive robustness

Figure S7 isolates the outer-envelope correction at fixed $\beta = -0.5$ and fixed 16 ns midpoint inversion. Both pulses use $L = 30$ μs, $c = 0.001$, and the same three-level coherence and anharmonicity parameters given above. The only change is replacement of the hard-truncated envelope by Eq. (S35), including both endpoint subtraction and the $p = 5$ window. The nominal peak scale is held fixed, without area rematching. These are the saved matched simulations used for the main-text pulse-improvement comparison, evaluated at 100 drive scales from 0 to 100 MHz and 201 detunings within ± 0.5 MHz.

The FWHM is extracted from the nearest half-depth crossings relative to a linear background fitted over $0.4 \leq |\Delta|/(2\pi) \leq 0.5$ MHz. The contrast C is the dip depth relative to this background, in excitation-probability units. To quantify visibility at a fixed shot budget, we additionally use the ideal SNR estimate from the saved analysis. Let p_c be the interpolated probability at the central dip and $p_\pm$ the probabilities at detunings ± 0.45 MHz. For $N = 1000$ independent binary measurements, with $N/2$

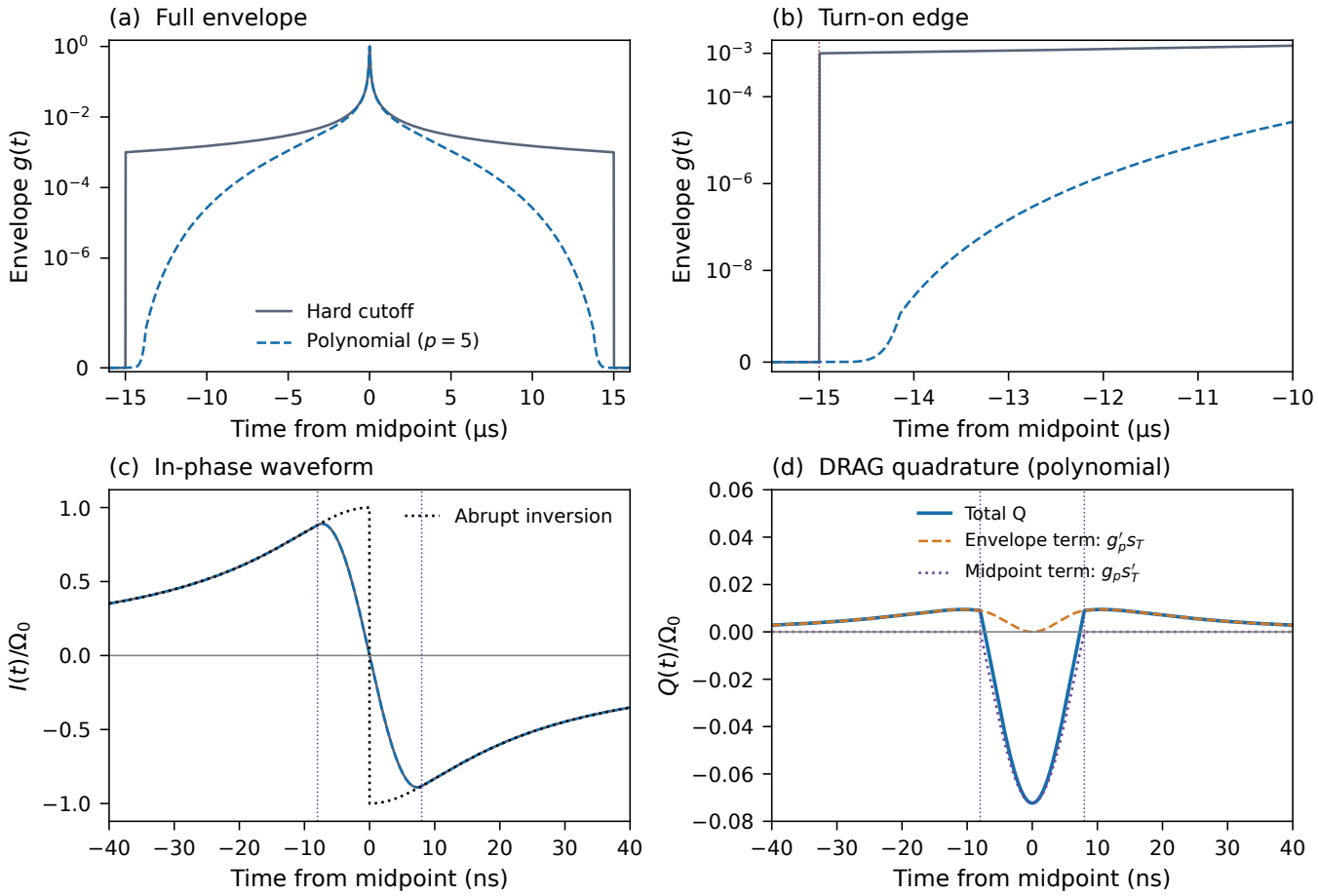


Figure S6. Envelope shaping and DRAG for $L = 30$ μs, $c = 0.001$, and a 16 ns midpoint transition. (a) Unsigned hard-cutoff and polynomial-corrected ($p = 5$) envelopes. (b) Turn-on edge; the turn-off is its mirror image. The envelope axes are logarithmic above 10^{-8} in (a) and 10^{-9} in (b), and linear below these values to include zero. (c) Signed in-phase waveform I/Ω_0 near the midpoint, including the abrupt hard-cutoff inversion as a reference. (d) Polynomial-pulse DRAG quadrature Q/Ω_0 for $\beta = -0.5$ and $\alpha/(2\pi) = -216$ MHz, decomposed into the envelope-derivative term $-\beta g_p' s_T/|\alpha|$ and midpoint term $-\beta g_p s_T'/|\alpha|$. Vertical dotted lines mark the 16 ns inversion interval in (c),(d). The convention is $I - iQ$; $g(0) = 1$ is held fixed without area or signed-peak renormalization.

shots at the center and $N/4$ at each shoulder,

$$\begin{aligned} C_{\mathrm{sh}} &= \frac{p_+ + p_-}{2} - p_c, \\ \sigma_C^2 &= \frac{2p_c(1-p_c) + p_+(1-p_+) + p_-(1-p_-)}{N}, \\ \mathrm{SNR}_{\mathrm{ideal}} &= \frac{C_{\mathrm{sh}}}{\sigma_C}. \end{aligned} \tag{S36}$$

Here C_{sh} uses two fixed shoulders, whereas the plotted C uses the fitted background. This SNR assumes perfect classification of P_e and excludes assignment error, technical noise, drift, and the measurement cost of locating the dip. It is a projection-noise estimate, not a measured experimental SNR.

At 100 MHz, edge correction reduces the FWHM from 506.0 to 161.8 kHz and raises C from 0.079 to 0.315. The ideal SNR increases from 2.50 to 10.64. The improvement persists across all 50 sampled drive settings from 50.51 to 100 MHz: the corrected pulse has worst-case FWHM 161.8 kHz and minimum ideal SNR 10.64, compared with 506.0 kHz and 2.50 for the hard cutoff. Thus the polynomial correction maintains a narrower, more visible dip throughout this high-drive interval. This is robustness to changing the nominal drive scale; no stochastic amplitude-noise process is simulated.

The roles of the two corrections remain distinct. In the same saved comparison, adding DRAG to the hard-cutoff pulse changes its 100 MHz FWHM from 484.8 to 506.0 kHz, so DRAG alone does not produce the narrowing. The edge-corrected width also remains above $1/(\pi T_2) = 43.5$ kHz. These results establish an improvement for the specified pulse, drive interval, and readout model, rather than a universal optimum or coherence-limited linewidth. The final section examines the polynomial family's behavior over the full cutoff and pulse-duration range.

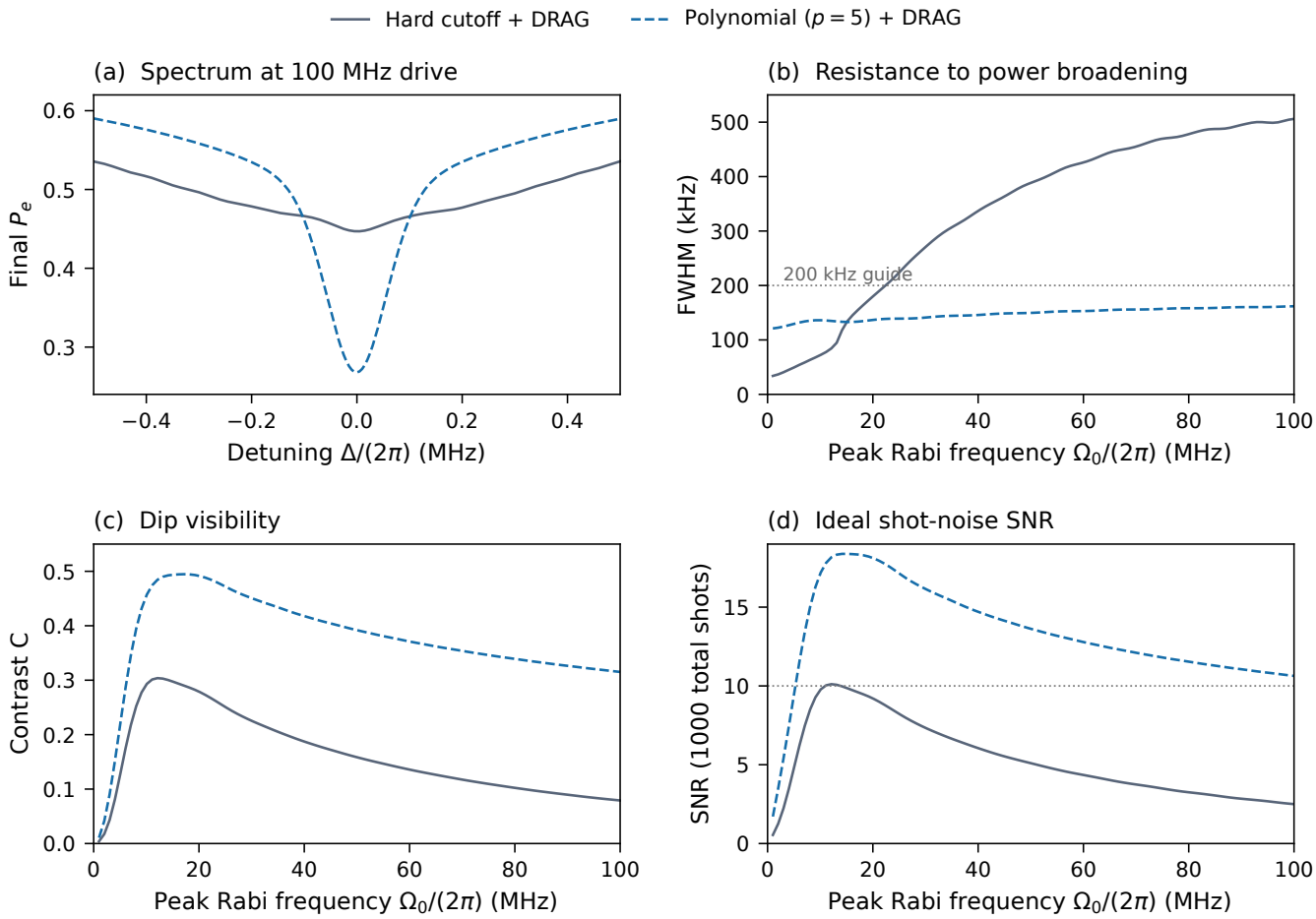


Figure S7. Effect of polynomial outer-edge smoothing at fixed DRAG. Gray solid curves show the hard cutoff; blue dashed curves add endpoint subtraction and the $p = 5$ window. Both use $L = 30$ μs, $c = 0.001$, $\beta = -0.5$, and a 16 ns sine midpoint transition. (a) Simulated spectra at $\Omega_0/(2\pi) = 100$ MHz. (b) Dip FWHM, (c) background-referenced contrast C, and (d) ideal SNR from Eq. (S36) versus peak drive. Dotted lines indicate illustrative criteria of FWHM ≤ 200 kHz and SNR ≥ 10. All 50 sampled polynomial-pulse points in the 50–100 MHz interval meet both criteria. Curves connect saved samples without additional smoothing or normalization; undefined zero-drive metrics are omitted.

S11. EXTENDED CUTOFF AND PULSE-DURATION RANGE

The duration–cutoff comparison in the main text displays the $10^{-5} \leq c \leq 10^{-2}$ portion of a larger numerical survey. Here we consider the full range, comprising 100 logarithmically spaced cutoffs from 10^{-5} to 1 and pulse durations $L = 10$, 20, 30, 45, and 60 μs. For every pair (L, c), we simulate 100 nominal drive scales $\Omega_0/(2\pi)$ from 0 to 100 MHz and 201 detunings $\Delta/(2\pi)$ from -0.5 to 0.5 MHz. This survey contains 50 000 spectra and 10 050 000 simulated operating points. It extends the parameter range of the three-level calculation, rather than the range of experimental validation. Figure S8 displays the full cutoff–drive maps for all five pulse lengths, with common color scales for linewidth and visibility.

S11.1. Pulse family and comparison metrics

We use the polynomial-corrected DRAG echo introduced in Sec. S10. The unsigned envelope is the cutoff-subtracted root-Lorentzian multiplied by the fixed polynomial window,

$$g(t) = \frac{[1 + (t/\sigma)^2]^{-1/2} - c}{1 - c} \left[1 - \left(\frac{2t}{L}\right)^2\right]^5, \qquad \sigma = \frac{L}{2} \frac{c}{\sqrt{1 - c^2}}, \tag{S37}$$

for $|t| < L/2$, with zero drive outside. The signed drive is $I(t) = \Omega_0 g(t) s(t)$, where $s(t)$ reverses sign through the same 16 ns sine transition at every operating point. The complex drive is $I - iQ$, with $Q = -\beta \dot{I}/|\alpha|$ and $\beta = -0.5$; α is an angular frequency and the deriva-

tive includes both g and s. The unsigned midpoint normalization is fixed, without pulse-area or smoothed-peak renormalization. The three-level Lindblad model uses $\alpha/(2\pi) = -216$ MHz, $T_1 = 51.24$ μs, and $T_2 = 7.31$ μs, with no added phenomenological Stark correction.

For this corrected pulse, c specifies the original root-Lorentzian core width; the actual endpoint amplitude vanishes for every cutoff. Thus the endpoint-jump estimate in Sec. S4 does not directly describe this family. Increasing L at fixed c also broadens σ, so this is not a sweep of tail duration at fixed core width. Moreover, the apparently singular endpoint $c = 1$ has the continuous limit

$$g(t)\big|_{c=1} = \left[1 - \left(\frac{2t}{L}\right)^2\right]^6 . \tag{S38}$$

It is therefore a smooth polynomial envelope, not a square pulse.

We extract a local dip FWHM Γ_f after subtracting a linear background fitted over $0.4 \le |\Delta/(2\pi)| \le 0.5$ MHz. The dip-center search is restricted to ± 0.25 MHz, and the width uses the nearest half-depth crossings. Widths spanning fewer than six detuning intervals (30 kHz) are rejected, as are unresolved dips and nonpositive shoulder contrast. Additional minima with population prominence at least 0.005 are counted but do not automatically invalidate a local width. Acceptance consequently does not certify a unique resonance.

Contrast is the height of a peak or depth of a dip relative to its local background, expressed as a probability difference on a scale from 0 to 1. For the simulated dips we use the same background as the FWHM extraction,

$$C = P_{\mathrm{bg}}(\delta_*) - P_e(\delta_*), \qquad \delta = \Delta/(2\pi), \tag{S39}$$

where δ_* is the sampled detuning of maximum background-relative depth within the central search interval. For a peak, the sign is reversed. A contrast of 0.3 therefore means a difference of 0.3 in excitation probability. Contrast is not divided by a noise estimate and is independent of the assumed number of shots. The reference $\Gamma_{f,T_2} = 1/(\pi T_2) = 43.5$ kHz provides a common linewidth scale, not an assertion that these shaped pulses attain that limit.

S11.2. Large-cutoff behavior and the duration tradeoff

The extended range shows why neither minimizing cutoff nor increasing it toward unity is sufficient to obtain useful resolution. Table S3 gives representative sampled cutoffs for $L = 30$ μs at $\Omega_0/(2\pi) = 49.495$ MHz. The smallest cutoff gives a narrower accepted dip but contrast of only 0.0027. Around $c = 10^{-3}$ the signal is much stronger; extending toward $c \simeq 0.1$ and 1 broadens the local feature while reducing its contrast. This high-cutoff degradation occurs despite zero endpoint amplitude and must be distinguished from abrupt-truncation artifacts.

Table S3. Representative points from the extended cutoff sweep at $L = 30$ μs and $\Omega_0/(2\pi) = 49.495$ MHz. Cutoffs are rounded saved samples; all listed widths pass the local-dip screen. Contrast follows Eq. (S39).

Cutoff c	FWHM (kHz)	Contrast C
10^{-5}	125.7	0.0027
1.048×10^{-3}	150.4	0.3878
9.545×10^{-3}	181.9	0.1895
9.770×10^{-2}	237.9	0.1093
1	325.9	0.0782

Table S4. Fixed-cutoff selections maximizing the worst sampled contrast/FWHM over the drive range. These are discrete survey results, without interpolation or a continuous optimization. Width and contrast extrema are evaluated separately at the selected cutoff.

L (μs)	Cutoff c	Max. FWHM (kHz)	Min. C	Min. C/Γ_f (MHz^{-1})
10	2.656×10^{-3}	362.5	0.574	1.740
20	1.177×10^{-3}	205.1	0.452	2.270
30	6.579×10^{-4}	154.9	0.321	2.329
45	3.678×10^{-4}	133.2	0.202	1.944
60	2.310×10^{-4}	145.4	0.123	1.371

Duration changes this balance. At the same drive and $c = 1.048 \times 10^{-3}$, increasing L from 10 to 45 μs reduces FWHM from 308.6 to 138.9 kHz while contrast falls from 0.723 to 0.220. At 60 μs, the width rises to 187.5 kHz and contrast falls to 0.114. At large cutoff, a narrower local feature can instead accompany nearly lost visibility: for $L = 45$ μs and $c = 1$, FWHM is 166.1 kHz but contrast is only 0.0117; the corresponding 60 μs spectrum fails the local-dip screen. These examples show that linewidth alone does not quantify usable sensitivity, and the duration dependence is not a universal $1/L$ narrowing law.

S11.3. One cutoff across the drive range

To quantify robustness without retuning cutoff at each drive, we select one cutoff per duration by maximizing the minimum C/Γ_f over all 90 sampled drive scales from 10.101 to 100 MHz. A candidate must have an accepted dip at every one of these scales. Table S4 reports the sampled selections; the worst width and minimum contrast need not occur at the same drive.

The selected cutoff decreases with duration. The 20 and 30 μs pulses have similar worst-case contrast/FWHM, with the latter trading some signal for

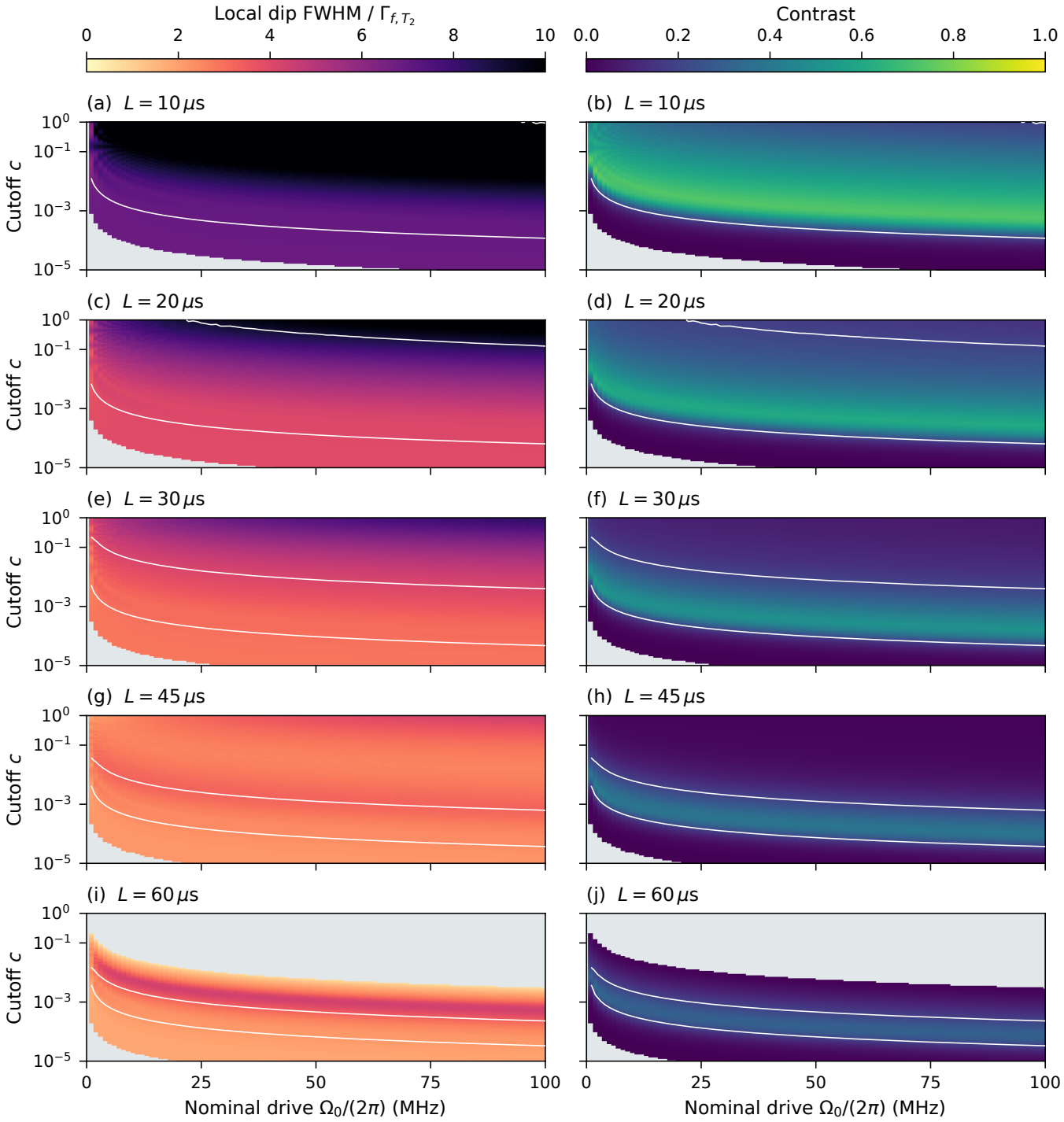


Figure S8. Full cutoff–duration survey for the polynomial-corrected DRAG echo. Rows show $L = 10$, 20, 30, 45, and 60 μs. Left: dip FWHM divided by $\Gamma_{f,T_2} = 43.5$ kHz; lighter colors indicate narrower linewidths, and values above 10 saturate at the darkest color. Right: probability contrast C [Eq. (S39)]. Thin white contours in both columns mark $C = 0.2$, interpolated between accepted samples. Each panel contains 100 logarithmic cutoffs and 100 drive scales spanning the displayed axes. Color scales are shared across rows; gray cells are rejected dips. All pulses use $p = 5$, $\beta = -0.5$, and a 16 ns sine inversion. Heatmaps show saved cells without interpolation.

a smaller worst-case width. The selected 10, 20, and 30 μs pulses maintain contrast above 0.3 throughout the sampled drive interval; the selected 45 and 60 μs pulses have minimum contrasts of 0.202 and 0.123, respectively. Even the smallest worst-case widths in this comparison remain about three times the T_2 reference linewidth. Contrast/FWHM is a heuristic tradeoff score rather than a frequency-estimation uncertainty. These selections do not establish a global waveform optimum or an experimentally calibrated operating point.

S11.4. Numerical resolution and scope

The integration uses fourth-order Runge–Kutta propagation with a maximum time step of 0.25 ns and local refinement near the core and midpoint transition. Representative convergence checks cover all five durations at $c = 10^{-5}$, 1.048×10^{-3}, and 1, and nominal drives of 10.101, 49.495, and 100 MHz (45 spectra). Halving the time step changes populations by at most 1.28×10^{-4} and accepted FWHM by 0.254 kHz. Doubling the detuning intervals changes accepted FWHM by at most 0.116 kHz. Neither refinement changes dip acceptance in these checks. Metric differences include only spectra accepted at both resolutions. These checks support the representative comparisons, including the $c = 1$ limit, but are not a full-grid error bound. The fixed detuning window and background convention also limit the interpretation of broad or structured spectra, independently of integration accuracy.

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